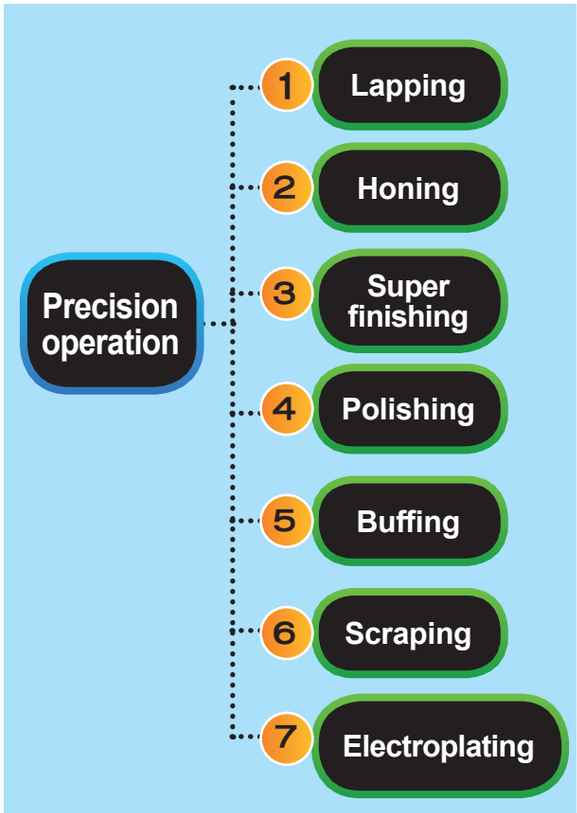


Lapping

Lapping is the abrading process that is used to produces geometrically true surfaces correct minor surface imperfections, improve dimensional accuracy to provide a very close fit between two surface in contact. Very thin layers of metal (0.005 to 0.01 mm) are removed in lapping. Machining can be done to the accuracy of 0.1 micron.



Lapping

To perform lapping operations, lapping shoes and lapping mixture are needed. Laps may be made of almost any material soft enough to receive and retain the abrasive grains. They are made of cast iron, Brass, Copper lead or soft steel. It is made in different shapes. Lapping mixtures may be made abrasive power such as emery, corundum, iron oxide, chromium oxide mixed with oil or special pastes.

The face of the lap becomes charged with abrasive particles. Laps may be operated by hands or by machine. Cylindrical work may be lapped by rotating the work in a lathe and reciprocating the lap over the work. Flat surfaces may be lapped by holding the work against a rotating disc. Special lapping machines like vertical lapping machine, centreless lapping machine and abrasive belt lapping machines are also widely used.

Honing

Honing is the abrading process done mostly for finishing round holed produced by drilling, reaming or boring by means of bonded abrasive stones called 'hones'. Honing is a machining process and it is used to remove metal upto 0.25 mm. The surface roughness value can be maintained between 0.025 and 0.4 microns. So honing is used to correct some out af roundness, tapers, tool marks and axial distortion.

Honing stones are used for performing honing tool head fitted with honing stones is fitted on spindle and rotates. The parts having holes to be honed are mounted on vises or suitable fixtures. The spindle is moved vertically to abrade the walls of the holes. A good quantity of coolant should be applied while honing.

XII BME_English version CHAPTER 04.indd 94 1/8/2020 1:43:08 PM